MIL-L-14519C
22 June 1978
SUPERSEDING
MIL-L-14519B
10 March 1972

MILITARY SPECIFICATION

LIMESTONE, PULVERIZED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pulverized limestone.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48	- Sacks, Shipping, Paper.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 536	- Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

D13216E5562 - Unitization of Bagged Commodities for Level A Shipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street NW, Washington, DC 20036.)

DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE (HEW)

U. S. Public Health Service Drinking Water Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime.
C 50 - Sampling, Inspection, Packing, and Marking of Lime and Limestone Products.
E 11 - Wire-cloth Sieves for Testing Purposes.

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(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The limestone shall conform to the chemical composition of Table I.

TABLE I. Chemical composition.

Constituent	Percent by weight	
	Maximum	Minimum
Silica (silicon dioxide, SiO_2)	2.0	-
Iron and aluminum oxides (Fe_2O_3 plus Al_2O_3)	1.0	-
Calcium carbonate ($CaCO_3$)1/	-	90.0
Magnesium oxide (MgO)	3.0	-
Loss of weight on ignition (CO_2)	45.0	-
Hygroscopic water (H_2O)	1.0	-

1/ Determined as calcium oxide (CaO).

3.2 Deleterious ingredients. When pulverized limestone in the amount of 400 milligrams (mg) is added to one liter of water (400 ppm suspension), the deleterious ingredients in the water shall not be increased by more than 60 percent of the amount established by the U. S. Public Health Service Drinking Water Standards. Where limits have been established for "Amounts not to Exceed" and "Cause for Rejection", the amounts established as "Cause for Rejection" shall be the governing factor.

3.3 Particle size. Not less than 90 percent of the limestone shall pass through a No. 100 sieve and not more than 35 percent shall pass through a No. 325 sieve.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for tests shall be in accordance with ASTM C 50.

4.3.2 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.1 through 4.4.1.3. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Tests. Unless otherwise specified (see 6.2), distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections shall be applied where significant. Results shall be reported to the nearest significant unit (see 6.3).

4.4.1.1 Chemical composition. The chemical composition shall be determined in accordance with ASTM C 25. Nonconformance to Table I shall constitute failure of this test.

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4.4.1.2 Deleterious ingredients. Place 400 mg of a composite specimen weighed to the nearest 5 mg in a glass beaker and add one liter of water measured to the nearest milliliter. Cover and set aside for 24 hours with occasional stirring by hand. At the end of 24 hours, filter the suspension through filter paper to remove all limestone particles. Test the filtrate for the deleterious ingredients specified in 3.2 in accordance with the methods outlined in Standard Methods for the Examination of Water and Wastewater. Deleterious ingredients in amounts exceeding limits specified in 3.2 shall constitute failure of this test (see 6.4).

4.4.1.3 Particle size. The particle size shall be determined in accordance with FED. TEST METHOD STD. No. 536, Method 2101, except that the sieves shall conform to ASTM E 11. Less than 90 percent of the limestone passing through a No. 100 sieve or more than 35 percent passing through No. 325 sieve shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a unitized load of completely filled bags prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 defects per 100 samples.

101.	Sack construction not as specified.
102.	Pallet not conforming to drawing, for Level A.
103.	Cap not conforming to drawing, for Level A.
104.	Banding not conforming to drawing, for Level A.
105.	Marking incomplete, missing, illegible, or incorrect.

5. PACKAGING

5.1 Packing. Packing shall be Level A or Commercial as specified (see 6.2 and 6.5).

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5.1.1 Level A. Limestone in unit quantities of 100 pounds (45.36 kilograms) shall be placed in shipping sacks conforming to UU-S-48, Type IV, Style B, Sack construction 9 or 9X, and then unitized in accordance with Drawing D13216E5562.

5.1.2 Commercial. Limestone in unit quantities of 100 pounds (45.36 kilograms) shall be packed in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The pulverized limestone is intended for use as a coagulant aid in military water purification equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Test mediums required, if other than specified (see 4.4.1).
(c) Degree of packing required (see 5.1).

6.3 Significant places. For the purpose of determining conformance to this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E 29).

6.4 Degree of packing (see 5.1). Only two levels of packing are suitable and adequate for this commodity. The sacks, specified in Level A, will also be used in the field as a storage container when a water purification equipment set is used. As such, it is susceptible to exposure to all environmental conditions in addition to rough handling. A lesser sack would require repacking prior to troop issue.

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Commercial packing is provided for the small amount of this commodity to be handled under Commercial conditions. The vast majority of all limestone will be used for field use and will thereby require Level A packing.

Custodians:
Army - ME
Air Force - 68

Review activity:
DLA - GS

User activity:
Navy - YD

Preparing activity:
Army - ME

Project 6810-B104

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-L-14510C Limestone, Pulverized

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

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